National Institute of Technology Kurukshetra

Knowledge Representation and Problem Searching-AIPC 201(AIML)

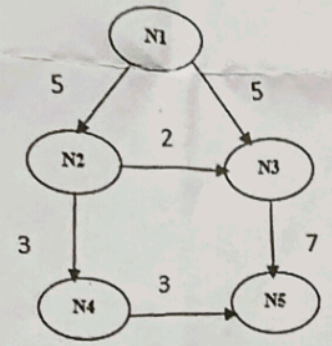
Mid-Term Examination-I, 3rd Semester, 2025-26

Time: 50 Minutes

Total Marks: 15

Note: Attempt all the questions. If require any missing data; then choose suitably.

Q1.A) Given a network with 5 nodes; N1, N2, N3, N4, N5, with the actual distance between the nodes written by the sides of the edges. N1 is the starting node and N5 is goal node. The heuristic value of each node is given in a table shown below. Use A* algorithm to find the path from N1 to N5 detailing the Frontier and Explored list in each step. **[3 marks]**



Node	H(Node)
N1	9
N2	5
N3	3
N4	1
N5	0

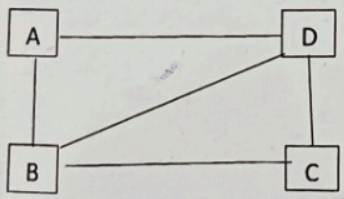
B) Verify with reason, if the given heuristic, shown in the table above is admissible. If not admissible heuristic, mention which nodes violate admissibility, do minimal changes to the heuristic table (change the heuristic value corresponding to the nodes that violate admissibility) to make it admissible. Also verify with reason, if the given heuristic is consistent. If not a consistent heuristic, mention which nodes violate consistency, do minimal changes to the heuristic table to make it consistent. **[3 marks]**

Q2. Given below, a graph with 4 nodes and 3 available colors (red, green, blue) to solve the graph coloring problem. In graph coloring problem, we have to assign the colors from the set of colors, to the nodes of the graph, so that, no adjacent vertices have same colors. The initial state S_0 is given as A=red, B=red, C=red & D=blue. It is also given that, A is always red and D is never green.

i) What is the number of neighbors of each state?

ii) Use BFS searching method to solve this graph coloring problem writing each step.

[2 + 2=4 marks]



Q3. A) Write and explain the model based basic entailment algorithm. **[3 marks]**

B) Given three Boolean variables X, Y, Z and Knowledge Base (KB), where KB is $(X \Leftrightarrow (Y \vee Z))$. The fact given is $\sim X$. Prove KB $\models \sim Z$, using inference rules and propositional logic. Explain each step. **[2 marks]**